

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – SCENARIO BASED

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO2 – Manipulation	Assessment:	Assessment Tool 2 – SCENARIO BASED
Weightage:	30% of CIA		
CO5 Statement:	Design inflectional, derivational, finite-state parsing, stemming algorithms and for analyse morphological methods. (BL-3)		
Student Name:	Harshavardhan RV	Reg. No.:	192472163
Section / Batch:	CSE - AI	Date:	1/8/2026

Code of Conduct

I Harshavardhan RV(192472163) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: HARSHAVARDHAN RV

Instructions

- Read all questions carefully before attempting the answers.
- Answer all five questions. Each question carries equal marks.
- Show all intermediate steps, calculations, probability computations, tagging decisions, and justifications wherever applicable.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
English Word Classes, Tagsets for English, Part of Speech Tagging	Morphological decomposition of analyzing, analysis, and analytical; identification of roots, affixes, inflectional and derivational forms, and normalization for indexing.	Q1
English Word Classes, Tagsets for English, Stochastic Part of Speech Tagging	Morphological parsing of disagree, agreement, and agreeable; identification of prefixes, suffixes, base forms, and semantic effects of derivation.	Q2
Counting Words, English Word Classes, Part of Speech Tagging	Morphology-based normalization of govern, government, and governance for topic modeling, clustering, and lexical standardization.	Q3
Rule-Based Part of Speech Tagging, Transformation-Based Tagging	Morphological analysis of activate, activation, and reactivation; identification of derivational layers, word-class changes, and semantic shifts.	Q4
Rule-Based Part of Speech Tagging, English Word Classes, Part of Speech Tagging	Inflectional normalization of create, creates, and creating; extraction of root forms and classification of tense/aspect transformations.	Q5

Annexure A – SCENARIO BASED

- 1. A large-scale search engine indexing system processes user queries containing word variants such as “analyzing”, “analysis”, and “analytical”. The system must improve retrieval accuracy by grouping semantically related terms under a unified representation. Design a morphological processing approach to decompose each input word into root and affixes, distinguish between inflectional and derivational forms, and normalize them for indexing. Apply the process to the given inputs and determine the root form along with the morphological structure of each word.

1. Search Engine Indexing System

Morphological Processing Approach

- ❖ Tokenize the input words.
- ❖ Identify prefixes and suffixes.
- ❖ Separate the root word from affixes.
- ❖ Distinguish **derivational** and **inflectional** morphology.
- ❖ Normalize all words to a common root (lemma) for indexing.

Morphological Analysis

Input Word	Root	Affix	Type	Morphological Structure	Normalized Form
analyzing	analyze	-ing	Inflectional (Present Participle)	analyze + ing	analyze
analysis	analyze	-sis	Derivational (Noun)	analyze + sis	analyze
analytical	analyze	-tic + -al	Derivational (Adjective)	analyze + tic + al	analyze

Result

- ❖ Common Root: **analyze**
- ❖ Search engine indexes all three words under **analyze**, improving retrieval accuracy.

2. An AI-based sentiment analysis platform receives product reviews containing words like “disagree”, “agreement”, and “agreeable”. The system must interpret sentiment consistently despite variations in word formation. Develop a morphological parsing strategy to identify prefixes, suffixes, and base forms, while capturing how derivational changes influence meaning. Process the given inputs to extract roots and classify each transformation, ensuring accurate semantic interpretation across all word forms.

2. AI-Based Sentiment Analysis Platform

Morphological Parsing Strategy

- ❖ Detect prefixes and suffixes.
- ❖ Extract the base/root word.
- ❖ Identify derivational transformations.
- ❖ Preserve semantic relationship during normalization.
- ❖ Map all forms to the common base.

Morphological Analysis

Input Word	Prefix	Root	Suffix	Transformation	Normalized Form
disagree	dis-	agree	—	Prefix (Negation)	agree
agreement	—	agree	-ment	Derivational (Noun)	agree
agreeable	—	agree	-able	Derivational (Adjective)	agree
-	-	-	-	-	-

Meaning Change

- ❖ **agree** → positive meaning
- ❖ **disagree** → opposite meaning due to prefix **dis-**
- ❖ **agreement** → state or act of agreeing
- ❖ **agreeable** → capable of being agreed with / pleasant

Result

All words are normalized to the base form **agree**, while derivational changes are retained for semantic interpretation.

3. A text analytics pipeline in a news aggregation system handles documents containing terms such as “govern”, “government”, and “governance”. The system must standardize these variations for topic modeling and clustering. Construct a morphology-based normalization mechanism that decomposes each word into its root and affixes, identifies derivational layers, and maps them to a common base form. Apply the approach to the input words and derive the normalized representation along with their morphological breakdown.

3. News Aggregation System

Morphology-Based Normalization Mechanism

- ❖ Identify the root word.
- ❖ Detect derivational suffixes.
- ❖ Remove derivational layers.
- ❖ Map every word to a unified base representation.
- ❖ Use normalized forms for clustering and topic modeling.

Morphological Analysis

Input Word	Root	Affix	Type	Morphological Structure	Normalized Form
govern	govern	—	Root	govern	govern
government	govern	-ment	Derivational (Noun)	govern + ment	govern
governance	govern	-ance	Derivational (Noun)	govern + ance	govern

Result

- ❖ Common Base Form: **govern**
- ❖ All words are indexed under **govern** for consistent topic modeling.

4. A document classification system processes input words such as “activate”, “activation”, and “reactivation” to improve semantic grouping and indexing. Construct a morphological parsing strategy to identify prefixes, suffixes, and root forms while distinguishing derivational layers. Analyze how prefix addition and suffix transformation affect meaning and word class. Apply the process to decompose each word and map them to a unified base representation. Derive the morphological structure and normalized root for each input word.

4. Document Classification System

Morphological Parsing Strategy

- ❖ Detect prefixes.
- ❖ Detect suffixes.
- ❖ Extract the root.
- ❖ Identify derivational layers.
- ❖ Normalize all words to a common base.

Morphological Analysis

Input Word	Prefix	Root	Suffix		Word Class	Normalized Form
activate	—	active	-ate		Verb	active
activation	—	active	-ate + -ion		Noun	active
reactivation	re-	active	-ate + -ion		Noun	active

Meaning Changes

- ❖ **activate** → make active.
- ❖ **activation** → process of making active.
- ❖ **reactivation** → activate again.

Result

All words are mapped to the normalized base **active**, enabling semantic grouping and document indexing.

5. A search optimization module processes input words such as “create”, “creates”, and “creating” to ensure consistent retrieval across grammatical variations. Design a morphology-based normalization approach focusing on inflectional transformations such as tense and aspect.

Identify suffix patterns and map all variations to a common base form without altering meaning.

Apply the process to extract root forms and classify each transformation.

Determine the morphological breakdown and normalized output for each word.

5. Search Optimization Module

Morphology-Based Normalization Approach

1. Identify inflectional suffixes.
2. Remove tense and aspect endings.
3. Preserve the original meaning.
4. Normalize every variation to the base form.
5. Store the normalized word for search indexing.

Morphological Analysis

Input Word	Root	Suffix	Transformation	Morphological Structure	Normalized Form
create	create	—	Base Form	create	create
creates	create	-s	Inflectional (Third Person Singular)	create + s	create
creating	create	-ing	Inflectional (Present Participle)	create + ing	create

Result

- ❖ Common Root: **create**
- ❖ All grammatical forms are indexed as **create**, ensuring consistent search results without changing meaning.

Rubrics for Calculation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Q. No. / Criteria Level	Understanding of Scenario	Identification of Key Issues	Application of Concepts	Decision Making	Clarity of Response	Sub. Total	Total %
1							
2							
3							
4							
5							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____